

Dexter Fernandes

London • +44 07436007369 • dexterfernandes11@gmail.com • linkedin.com/in/dexterferns •
<https://github.com/Dexter-Fernandes>

Computer Vision Engineer

Computer Vision Engineer with 3+ years commercial experience designing, training, and deploying production-grade ML/CV models in real-time, safety-critical environments. Hands-on with PyTorch and TensorFlow across the full model lifecycle including production monitoring in the cloud deployment on Microsoft Azure. Strong MLOps practice spanning CI/CD, MLflow, and DVC.

WORK EXPERIENCE

GrowthStage • Remote

05/2026 - Present

Recruiter

- Evaluated computer vision candidates on their technical depth across model development, deployment, and MLOps practices, maintaining direct exposure to current production engineering standards.

Esbaar • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Owned the end-to-end computer vision pipeline for real-time perception in industrial environments, improving F1 by 15% while maintaining deterministic 24/7 operation on British Petroleum rigs.
- Built multimodal data ingestion pipelines across image and sensor data using Python, Pandas, Airflow, MLflow, DVC, Optuna, and CVAT, processing 3.6 million records.
- Hardened deployed AI services with Dockerised microservices, Kafka messaging, and structured logging, reducing live video pipeline crashes by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, improving maintainability and reducing defects.
- Implemented CI/CD validation workflows for edge inference modules, improving release consistency across hardware targets for 24/7 deployments.
- Owned production monitoring and post-deployment validation for deployed vision systems, using observability tooling, failure analysis, and drift tracking to diagnose issues early and improve reliability in live environments.

LivNSense Technologies • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI

- Led a team to design, train, and deploy production object detection models for industrial safety in TensorFlow, reducing false positives by 33% for Austin Industries and John Deere.
- Built and maintained large-scale image and sensor datasets, automating labelling, versioning, and performance tracking to support anomaly investigation and continuous improvement cycles.
- Used confusion-matrix failure clustering to diagnose recurring models failure, to prioritise dataset fixes, and improve model robustness across camera feeds.
- Evaluated edge versus cloud inference trade-offs to minimise bandwidth, latency, and operational cost while maintaining safety-critical performance, supporting production cloud-hosted ML workloads on Microsoft Azure.
- Deployed quantised vision models using FP16/INT8 optimisation and pruning and layer fusion while meeting production accuracy targets.
- Built automated labelling and versioning workflows, cutting manual labelling effort by 80% while accelerating iteration cycles on large production datasets, reducing model rollout time from 72 hours to under 24.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

Dissertation: Radiological SLAM with LiDAR Odometry

- Radiological SLAM with LiDAR Odometry. Designed and implemented a real-time SLAM system using RTAB-Map on a Clearpath Husky, integrating LiDAR and radiological data for localisation and mapping in inspection scenarios.
- Configured Ceres, g2o, and GTSAM as solver backends and tuned ICP parameters across environments.
- Evaluated RTAB-Map, LIO-SAM, and Cartographer across comparable scenarios using ROS2 and rosbag2-based workflows to assess mapping and localisation behaviour.
- Validated sensor placement for coverage and calibrated LiDAR extrinsics to the robot base, including the Velodyne VLP-16 and Intel RealSense D435i.
- Implemented a ROS2 (rclcpp) point cloud pipeline in C++17 using PCL, with tf2-based frame management and message synchronisation to resolve transform timing issues.

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

- Built a strong foundation in signals, systems, embedded hardware, and control, which later supported work in real-time AI, sensor systems, and robotics software.

PROJECTS

DeepStream RTSP Pipeline

06/2026

- Built an end-to-end NVIDIA DeepStream 9.0/GStreamer RTSP video analytics pipeline for multi-stream computer vision, using rtspsrc, nvstreammux, nvinfer, nvtracker, and nvdsosd to run YOLO+NvDCF tracking, metadata extraction, anonymisation, CSV logging, and re-streamed output across 3 concurrent camera feeds.
- Deployed and optimised a YOLO26n TensorRT FP16 detector on 6GB VRAM Turing hardware, combining dynamic-batch engine conversion, nvinfer integration, p99 latency profiling, and FP16-vs-FP32 accuracy validation to sustain 29.7 FPS per stream live and 131.3 FPS per stream unthrottled.
- Implemented a C++/CUDA TensorRT decode plugin using IPluginV2DynamicExt, moving post-processing into the inference engine to support custom-layer deployment.
- Added production hardening through GPU smoke tests, model-promotion gates, structured JSON logging, and per-camera health metrics for reactive debugging across liveness, dropped frames, reconnect events, and RSS.

SKILLS

- **Programming & Tooling:** Bash, C++17, CUDA C++, Git, Linux, NumPy, OpenCV, Pandas, PIL, PostgreSQL, Python, PyTorch, Scikit-image, Scikit-Learn, TensorFlow
- **Cloud, Edge & MLOps:** Apache Airflow, Apache Kafka, API design, CI/CD, Docker, DVC (Data Version Control), Microsoft Azure, MLflow, Production monitoring, Structured logging
- **Edge Inference and Optimisation:** Convolutional Neural Networks, dynamic batching, FP16/INT8 deployment, Layer fusion, ONNX, OpenVINO, Optuna, TensorRT
- **AI-Assisted Development:** ChatGPT Codex, Claude Code
- **Computer Vision Models:** ByteTrack, EfficientDet, Instance Segmentation, NvDCF, NvSort, Object detection, Object tracking, RT-DETR, YOLO
- **Datasets, Annotation & Evaluation:** Confusion-matrix-driven failure clustering, CVAT, Dataset versioning, GRAD-CAM, Synthetic data generation, TIDE (Toolbox for Identifying Detection Errors)